SMART INDIA HACKATHON 2026

National Technical Research Organisation (NTRO)

Satellite-Based Oil Spill Detection, Drift Hindcasting & Vessel Attribution

Research Report + Product Requirements Document

Project codename: ***OilTrace***

Problem video:

AIS sample source:

Executive Summary

Marine oil spills can be visible from space but remain difficult to attribute to a vessel because three uncertain systems must be joined correctly: image interpretation, ocean transport, and incomplete vessel telemetry. Team Revolution proposes OilTrace, an analyst-in-the-loop platform that detects candidate slicks in Synthetic Aperture Radar (SAR) imagery, characterises their geometry, estimates a probability distribution over possible release locations and times, forecasts likely movement, and ranks vessels whose historical AIS trajectories are consistent with that evidence.

The project deliberately avoids three weak claims. First, a dark SAR patch is not automatically oil; low wind, biogenic films, internal waves, rain cells, wakes, and processing artefacts can look similar. Second, a single image cannot reliably yield exact slick thickness or age. Third, AIS correlation cannot establish legal culpability: gaps can arise from coverage and data-access limitations as well as deliberate behaviour. The product therefore returns *candidate detections*, *credible origin regions*, and *ranked investigative leads*, with confidence and provenance attached to every output.

The committed SIH minimum viable product (MVP) uses Sentinel-1 Level-1 GRD imagery, a U-Net segmentation baseline trained with oil and look-alike labels, ERA5 atmospheric forcing, Copernicus Marine or suitable INCOIS ocean-current fields, an ensemble Lagrangian drift model through OpenDrift/OpenOil, and historical or synthetic AIS stored as Parquet and queried with DuckDB. The interface overlays the slick polygon, backward-origin contours, forward-drift envelope, and reconstructed vessel tracks on one map. A scorecard explains each candidate using spatial consistency, temporal consistency, release-to-observation simulation fit, AIS reliability, and behavioural context. Scores are not labelled probabilities unless calibrated on labeled end-to-end cases.

The scientific foundation is established by operational and peer-reviewed precedent. EMSA's CleanSeaNet combines SAR analysis with meteorological, oceanographic, AIS, and vessel-detection information to support polluter identification [1]. NOAA's GNOME family models oil transport and weathering using wind and current forcing [2]. OpenDrift supports ensemble trajectory modelling and backward runs [3], [4]. Published work has already shown SAR–AIS–environmental fusion for tracing suspected spill vessels [5]; consequently, OilTrace's credible innovation is not the mere coexistence of those inputs. It is the open, reproducible, uncertainty-aware integration, the explicit "insufficient evidence" state, the evidence ledger, and evaluation of ranking calibration.

Decision in one sentence

Build the smallest end-to-end chain that can be scientifically tested: detect a slick, infer an uncertain source region/time, correlate complete and incomplete AIS tracks, rank candidates transparently, and let an analyst approve or reject the lead.

Document map

Part	Purpose	Primary outputs
Phase I: Research	Establish physics, datasets, prior art, scientific assumptions, uncertainty, and validation design.	Research questions, method, equations, dataset audit, metrics, failure modes.
Phase II: PRD	Convert the research decisions into buildable scope and acceptance criteria.	Requirements, architecture, user flow, APIs, UI, security, tests, roadmap, ownership.

Contents

Acronyms and notation

AIS	Automatic Identification System	AOI	Area of interest
ADE/FDE	Average/final displacement error	CMEMS	Copernicus Marine Service
EO	Earth observation	ECE	Expected calibration error
GRD	Ground Range Detected	H3	Hierarchical hexagonal geospatial index
IoU	Intersection over Union	MMSI	Maritime Mobile Service Identity
NFR	Non-functional requirement	PRD	Product Requirements Document
SAR	Synthetic Aperture Radar	SOG/COG	Speed/course over ground

PHASE 1

Research and Scientific Foundation

Phase purpose

Establish what is scientifically defensible, what data are actually available, how uncertainty enters the result, and which claims the prototype can validate.

1 Problem Definition and Research Questions

1.1 Problem statement

The NTRO challenge asks for an automated pipeline that: (a) detects and characterises an oil slick and, where feasible, estimates its age; (b) uses oceanographic and meteorological data to trace the slick backward toward a probable origin and forward toward affected areas; and (c) reconstructs historical vessel traffic from AIS, filters irrelevant traffic, and ranks plausible source vessels using proximity, trajectory and behavioural evidence.

1.2 Research questions

RQ1: Can a compact segmentation model distinguish oil candidates from look-alikes on unseen SAR scenes with acceptable precision and recall?
RQ2: Can ensemble transport inference produce origin contours that contain a known or simulated release point at the stated coverage rate?
RQ3: Does drift-aware spatio-temporal matching rank the true synthetic culprit above proximity-only baselines?
RQ4: Can every ranked candidate be explained using reproducible evidence and data-quality indicators?
RQ5: Under what conditions must the system abstain and return "insufficient evidence"?

1.3 Success definition

Success is an end-to-end, reproducible demonstration in which a detected slick is transformed into a georeferenced investigation case; the known synthetic source is inside the reported origin contour; the planted culprit appears in the top-ranked candidates; and the UI exposes enough evidence for an analyst to audit or reject the conclusion. A visually impressive map without component metrics is not sufficient.

2 Scope, Assumptions and Non-Goals

Area	In scope for SIH MVP	Explicitly out of scope / deferred
Detection	Sentinel-1 GRD candidate segmentation; oil/look-alike discrimination; polygon geometry.	Guaranteed oil chemistry, volume or thickness from a single SAR scene.
Transport	Ensemble backward transport and forward forecast with environmental perturbations.	Exact deterministic source point; physically reversing weathering, diffusion or emulsification.
Attribution	Historical AIS cleaning, short-gap reconstruction, candidate filtering and explainable scoring.	Declaration of guilt; identity resolution for vessels never observed by AIS.
Operations	Single-region batch case creation; downloadable evidence report; audit log.	Global continuous surveillance, commercial SAR tasking, or production enforcement action.
Data	Open imagery, public MetOcean products, MarineCadastre sample or synthetic AIS.	Assumption that US public AIS represents Indian operational availability.

Scientific guardrails

Thickness is not reported from SAR alone. Age is an interval supported by transport scenarios, not a direct image measurement. Backward modelling is probabilistic inference: advection may be integrated backward, but stochastic diffusion and oil weathering are not time-reversible. AIS gaps reduce data reliability; they do not prove intent.

3 Scientific Basis

3.1 Why SAR is the primary sensor

Surface oil damps capillary and short gravity waves, lowering radar backscatter and often creating a dark region in SAR imagery. C-band Sentinel-1 supports day/night, cloud-independent maritime observation; its GRD products are detected, multi-looked and projected to ground range [6], [7]. SAR is therefore the primary detector, while optical imagery is optional corroboration under daylight and sufficiently clear conditions.

Dark appearance is necessary but not sufficient. Established reviews emphasize segmentation, feature extraction and classification against look-alikes [8], [9]. The model must therefore learn or explicitly use context such as local contrast, texture, wind regime, shoreline proximity, shape, orientation and temporal persistence.

3.2 Supportable slick characterisation

For a georeferenced polygon Ω, the pipeline reports area $A(\Omega)$, perimeter $P(\Omega)$, centroid, principal orientation, major/minor axis, elongation, compactness $4\pi A/P^2$, fragmentation count and mask uncertainty. These are traceable to the segmented geometry. Oil type, volume and thickness remain "unknown" unless corroborating sensor or field data exist.

3.3 Transport model

For ensemble member k, a surface element position x_k is evolved as

$$\frac{d\mathrm{x}_k}{dt} = \mathrm{u}_c(\mathrm{x}_k, t) + \beta_k \mathrm{u}_{10}(\mathrm{x}_k, t) + \mathrm{u}_s(\mathrm{x}_k, t) + \boldsymbol{\eta}_k(t), \tag{1}$$

where u_c is ocean current, u_{10} is 10-m wind, β_k is an uncertain windage term, u_s is Stokes drift where available, and $\boldsymbol{\eta}_k$ represents unresolved dispersion. OpenDrift is a documented open framework for such particle trajectories, accepts multiple environmental readers, supports ensembles and permits negative time steps for backward runs [3], [4]; OpenOil adds oil-specific transport and fate processes [10].

For hindcasting, weathering is not "un-weathered." Instead, multiple plausible release hypotheses are propagated to or from the observed mask, and their agreement is used to construct an origin distribution. NOAA's WebGNOME and PyGNOME provide a second reference implementation for oil transport, weathering and uncertainty-aware trajectory work [2], [11].

3.4 Origin distribution

Let $\mathrm{x}_k(\tau)$ be a particle location at candidate release time $\tau < t_{obs}$. On grid cell g,

$$P_{origin}(g, \tau \mid S, M) = \frac{\sum_{k=1}^{N} w_k \mathbb{1}[\mathrm{x}_k(\tau) \in g]}{\sum_{k=1}^{N} w_k}, \tag{2}$$

where S is the observed slick and M denotes forcing and parameter scenarios. Report 50% and 90% highest-density contours, release-time weights, and sensitivity to forcing products. These contours are credible regions of the chosen ensemble, not universal guarantees.

3.5 AIS evidence

AIS provides identity, position, time, SOG, COG and related navigation messages for covered vessels. IMO carriage requirements apply to specified ship classes, not every craft, and web publication or access may carry safety and security considerations [12]. MarineCadastre provides a public US coastal data route, but its ordering service status and geographic coverage must be checked at demo time [13]. Indian operational deployment requires authorised AIS access or a validated alternative.

For vessel v with reconstructed path $\mathrm{z}_v(t)$, candidate generation first applies hard data-quality and time-window filters. Short gaps may be interpolated only when implied speed and turn rate remain feasible; long gaps remain explicit gaps. A gap is a reliability feature, never direct evidence of intent.

4 Data and Dataset Audit

Table 1: Data sources selected for the prototype and their constraints.

Source	Modality / role	Access	What it supports	Limitation / control
Sentinel-1 GRD	C-band SAR imagery	Copernicus Data Space	Primary slick candidate detection; VV/VH where available; calibrated backscatter pipeline.	Pixel spacing is not identical to effective resolution; orbit/-mode/polarisation are recorded per scene.
Sentinel-2	Multispectral optical	Copernicus Data Space	Optional corroboration and contextual view.	Cloud, sunglint, haze, daylight and acquisition-time mismatch.
Trujillo-Acatitla Part I–III	Sentinel-1 Sigma0 VV/VH with masks	Zenodo DOI set	Train/validation/test material for oil and look-alike segmentation [14]–[16].	Large downloads; geography and label protocol must be documented in the experiment card.
Krestenitis et al.	Sentinel-1 semantic segmentation	Public research dataset	Five-class benchmark including oil, look-alike, ship, land and sea [17].	Fixed historic dataset; domain shift to Indian waters must be tested.
ERA5	Atmospheric reanalysis	Copernicus Climate Data Store	10-m winds and wave-related atmospheric context.	Not the selected primary ocean-current source.

Source	Modality / role	Access	What it supports	Limitation / control
Copernicus Marine	Ocean analysis/forecast	Global Physics Analysis and Forecast product	Hourly surface currents; global 1/12-degree product and forecast metadata [18].	Coarse relative to ports/coasts; compare with regional/INCOIS data where available.
INCOIS OOSA/OSF	Indian operational guidance	Government of India service	India-specific trajectory and ocean-state reference; OOSA reports forecasts up to 96 h [19].	Data/API availability and licensing must be confirmed for machine ingestion.
MarineCadastre AIS	Terrestrial AIS, US	Bulk public downloads	Reproducible real-track demo and schema validation.	US-only evidence; AccessAIS order service may be unavailable while bulk files remain accessible.
Synthetic AIS benchmark	Generated tracks with ground truth	Project-owned	End-to-end attribution metrics, controlled gaps, dense traffic and injected culprit.	Demonstrates algorithm behaviour, not operational accuracy.

4.1 Minimum data contract

Every case stores: source URI and checksum; acquisition time in UTC; CRS; SAR mode/polarisations; preprocessing parameters; model version; forcing product/version and timestamps; particle seed; AIS source and coverage interval; quality flags; and output artifact hashes. Missing metadata blocks case promotion to "review-ready."

5 Prior Art, Gap and Positioning

EMSA CleanSeaNet is operational proof that SAR, meteorological/oceanographic context, AIS and vessel detection can support pollution monitoring and possible source identification; official material notes that trained operators assess these inputs and alerts can be issued rapidly after acquisition [1]. NOAA GNOME provides public oil trajectory and fate modelling [2]. OpenDrift provides an open programmable trajectory framework [3]. Deep-learning work demonstrates semantic segmentation and large-scale detection on SAR scenes [17], [20].

Importantly, Luo et al. already report a multi-source method combining SAR, AIS, wind, currents and bidirectional drift to trace suspected oil-spill vessels [5]. Therefore, a claim that "nobody has combined SAR and AIS" would be false.

Capability	CleanSeaNet	GNOME	Published fusion	OilTrace focus	Research gap addressed
SAR candidate detection	✓	–	✓	✓	Reproducible open baseline and label audit
Backward/forward transport	Contextual	✓	✓	✓	Ensemble contours and sensitivity record
AIS candidate correlation	✓	–	✓	✓	Explicit data-quality-aware filtering
Candidate explanations	Analyst-led	–	Limited	✓	Component scorecard and evidence ledger
Calibrated abstention	Not public	–	Not established	Target	"Insufficient evidence" acceptance tests
End-to-end open benchmark	Restricted service	Tool only	Case study	Target	Synthetic plus component-level real validation

Table 2: Conservative novelty positioning; "not public" is not treated as "does not exist."

6 Proposed Research Method

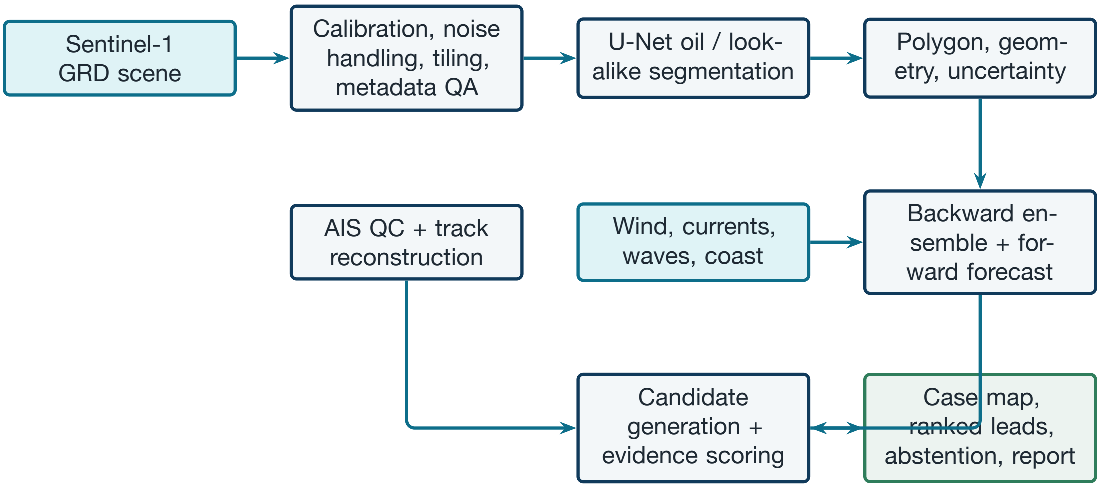

Figure 1: Research pipeline. All connectors are orthogonal; each stage emits versioned artifacts and quality flags.

6.1 Experiment stages

1. Detection baseline. Train U-Net with class-balanced patch sampling and group splits by source scene/event to prevent tile leakage. Compare against a deterministic dark-patch baseline and, if time permits, a SegFormer comparison.
2. Uncertainty. Use validation-calibrated thresholds plus either a small model ensemble or test-time augmentation. Report pixel confidence, scene quality and excluded conditions separately.
3. Geometry. Vectorise masks, preserve holes and fragments, remove only thresholds fixed on validation data, and compute geographic area in an appropriate projected CRS.
4. Hindcast/forecast. Seed particles over the whole slick polygon; perturb windage, current, diffusion, release time and initial location. Produce density rasters and 50/90% contours for multiple look-back horizons.
5. AIS reconstruction. Deduplicate, validate timestamps and coordinates, segment tracks at long gaps, and interpolate only short feasible intervals. Generate features at each origin-time slice.
6. Ranking. Compare a proximity baseline with the full physics-consistency score. Calibrate only if sufficient labeled cases exist; otherwise call the result an index on a 0–100 scale.
7. Ablation. Measure performance without drift, without behavioural context, with degraded AIS, and with alternate forcing to expose where value and fragility arise.

7 Attribution Model

7.1 Candidate generation

For every time slice τ in the release window, candidate vessel v is retained when its quality-controlled track intersects a buffered non-negligible portion of $P_{origin}(\cdot, \tau)$. Traffic outside the case bounds, time window, and navigational feasibility constraints is discarded before ranking.

7.2 Evidence features

Feature	Symbol	Definition	Interpretation control
Spatial consistency	s_{space}	Maximum or integrated origin density along the vessel path.	Primary physical evidence; robust to one noisy ping.
Temporal consistency	s_{time}	Alignment with the release-time distribution.	Broadens when age is uncertain.
Forward-fit	s_{fit}	Similarity of a forward release from vessel path to the observed slick.	Prevents backward-only false matches.
Track continuity	q_{AIS}	Coverage, gaps, impossible jumps and identity consistency.	Reliability multiplier, not guilt evidence.
Behaviour context	s_{beh}	Slowdown, turn-rate change, loitering relative to local/own baseline.	Weak supporting signal; vessel type is not decisive.
Data coverage	q_{case}	Imagery, forcing and AIS completeness.	Controls abstention and displayed confidence.

An initial transparent score is

$$R(v) = 100\, q_{case}\, q_{AIS}(v)\,(0.35 s_{space} + 0.25 s_{time} + 0.25 s_{fit} + 0.15 s_{beh})\,, \tag{3}$$

with weights frozen before the held-out evaluation. They are design defaults, not learned truth. If calibration data become available, a monotonic calibration layer maps R to an empirical probability; until then the UI shows evidence score, not "likelihood."

7.3 Abstention

The engine returns "insufficient evidence" when any mandatory input fails, all candidates have low physical consistency, the top candidates are statistically indistinguishable, or case uncertainty exceeds the configured threshold. This is a required product outcome, not an error.

8 Evaluation and Verification Plan

Table 3: Metrics and claim boundaries.

Layer	Metrics	Evaluation design	Claim allowed
Segmentation	Per-class IoU, Dice/F1, precision, recall, PR-AUC; scene-level false alarms	Event/scene-held-out split; separate geography and wind bins; no overlapping tiles across splits	"Detects candidates on this benchmark"
Geometry	Area/centroid error, boundary F1, Hausdorff distance	Compare vector outputs with held-out masks	"Measures visible mask geometry"
Hindcast	Source geodesic error, 50/90% contour coverage, contour area/sharpness	Synthetic known releases; real drifter or documented cases where available	"Recovers source under tested forcing"
Forecast	ADE/FDE, observed-slick overlap, shoreline-arrival interval	Multi-temporal observations or controlled simulations	"Forecast skill for tested horizons"
Attribution	Top-1, Top-3, MRR, no-candidate recall	Synthetic traffic with known culprit; dense and sparse scenarios	"Ranking works on benchmark scenarios"
Calibration	Brier score, ECE, reliability curve	Only when enough independent labeled cases exist	Probability terminology permitted only after pass
Operations	End-to-end latency, job failure rate, evidence completeness, analyst review time	Repeated demo runs from clean environment	"Prototype is reproducible"

8.1 Failure and adversarial test matrix

Test	Expected behaviour	Pass condition
Low-wind look-alike	Lower detection confidence; show wind context; permit rejection.	No forced vessel attribution.
Dense traffic	Rank several candidates; widen uncertainty.	Culprit in Top-3 or abstention; no fabricated certainty.
AIS reception gap	Preserve gap; avoid long interpolation.	Reliability flag visible in UI/report.
Spoofed/impossible jump	Split/flag track.	Invalid segment excluded from physical overlap.
Wrong current field	Sensitivity run diverges.	Case confidence falls and alternate forcing is logged.
No vessel in coverage	Return no-candidate state.	No placeholder suspect created.
Two plausible sources	Preserve multimodal origin field.	UI shows both modes; scoring does not collapse them silently.
Model domain shift	Flag out-of-distribution scene or weak quality.	Analyst review required before export.

9 Research Conclusions and Product Decisions

Decisions frozen for the SIH build

- U-Net is the MVP detector; SegFormer is a comparison, not a dependency. SAM and graph neural networks are future research.
- Sentinel-1 is primary; optical data are optional corroboration.
- ERA5 supplies atmospheric forcing; Copernicus Marine or suitable INCOIS products supply currents.
- DuckDB + Parquet is the prototype store; PostGIS is the scale-up path.
- The score remains an evidence index until empirical calibration is demonstrated.
- The system can abstain. Every export is a lead-generation report, not a legal conclusion.

PHASE 2

Product Requirements Document

Phase purpose

Translate the research decisions into implementable requirements, interfaces, tests, responsibilities, and an evidence-led SIH demonstration.

10 Product Vision and Stakeholders

Vision: Reduce the time from satellite observation to a reviewable, evidence-backed list of candidate vessels and predicted impact zones, while making uncertainty impossible to hide.

	Persona	Goal	Product need
Primary users	Maritime intelligence analyst	Determine which vessel tracks are consistent with an observed slick.	Unified time-controlled map, score decomposition, AIS quality, evidence export.
	Pollution response officer	Understand where the slick may move and which assets/coasts are at risk.	Forward envelope, arrival windows, downloadable geospatial layers.
	Remote-sensing analyst	Confirm or reject the automated slick mask.	Source imagery, preprocessing metadata, editable review status and look-alike evidence.
	System administrator	Keep inputs, models, jobs and access trustworthy.	Health, lineage, audit trail, role controls and reproducible re-runs.

10.1 User stories

- As a remote-sensing analyst, I want to inspect the raw and processed SAR scene beside the candidate mask so that I can reject a look-alike before expensive modelling.
- As a maritime analyst, I want candidate scores decomposed by space, time, forward-fit and track quality so that I can challenge the ranking.
- As a response officer, I want 6/12/24/48-hour forward envelopes so that I can prioritise shoreline and asset protection.
- As an administrator, I want every result tied to input checksums, model versions and parameters so that the case can be reproduced.

11 End-to-End User Flow

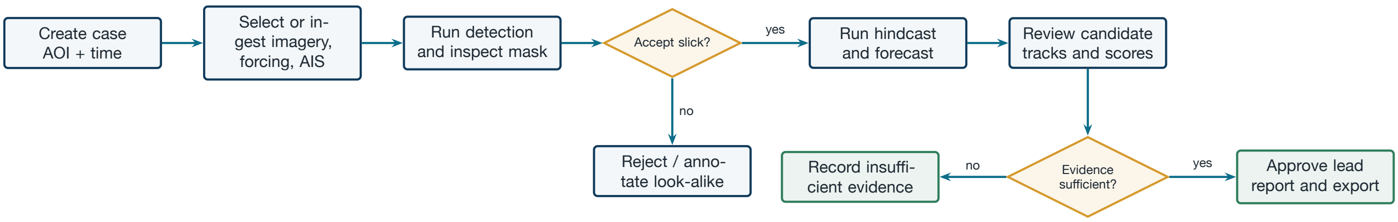

Figure 2: Operator workflow with two legitimate terminal states: approved investigative lead or insufficient evidence.

12 Scope and Prioritisation

Priority	Capability
MUST	Case creation; Sentinel-1 ingest; U-Net mask; analyst accept/reject; geometry; ensemble hindcast; forward envelope; AIS QC/reconstruction; candidate scorecard; abstention; map; evidence export; audit log.
SHOULD	Optical corroboration; alternate forcing sensitivity; time slider; model/data quality dashboard; baseline comparisons; GeoJSON/CSV export.
COULD	SegFormer comparison; H3 acceleration; scheduled scene discovery; shoreline exposure layer; multi-user comments.
WON'T	Exact oil thickness/volume; fully autonomous legal attribution; live commercial satellite tasking; dark-vessel identification without another sensor; GNN/SAM production dependency.

13 Functional Requirements

Table 4: MVP functional requirements and acceptance criteria.

ID	Requirement	Acceptance criterion	Priority
FR-01	Create investigation case	User can enter AOI/time or select a stored demo; case receives immutable ID and UTC timestamps.	Must
FR-02	Ingest SAR scene	System stores source URI/checksum, mode, orbit, polarisation, CRS and acquisition time; missing mandatory metadata blocks processing.	Must

ID	Requirement	Acceptance criterion	Priority
FR-03	Preprocess imagery	Calibration/noise/tiling configuration is versioned and output is georeferenced.	Must
FR-04	Detect candidate slick	Service emits probability raster, class mask and quality summary; inference can be rerun by model version.	Must
FR-05	Human detection review	Analyst can accept, reject or mark uncertain and record a reason; rejected cases cannot auto-trigger attribution.	Must
FR-06	Characterise geometry	Accepted mask produces valid GeoJSON and area, perimeter, centroid, orientation, axes, compactness and fragments.	Must
FR-07	Load environmental forcing	Wind and current coverage is checked for AOI/time; product/version and missing cells are displayed.	Must
FR-08	Run hindcast	Configurable ensemble emits time-indexed density and 50/90% origin contours with parameter manifest.	Must
FR-09	Run forward forecast	Service emits 6/12/24/48-hour envelopes where forcing is available and labels horizon validity.	Must
FR-10	Ingest AIS	Parser validates schema, normalises UTC, removes duplicates and flags invalid positions/jumps.	Must
FR-11	Reconstruct tracks	Only gaps below configured feasibility thresholds are interpolated; original and derived points remain distinguishable.	Must
FR-12	Generate candidates	Vessels outside origin/time/quality gates are excluded with reason codes.	Must
FR-13	Score candidates	Each candidate has total score, component values, data-quality multiplier and supporting timestamps/geometries.	Must
FR-14	Abstain	Engine returns a machine-readable no-candidate/insufficient-evidence reason when gates fail.	Must
FR-15	Visualise evidence	Map toggles source image, mask, origin contours, drift envelopes, candidate tracks and timestamps without page reload.	Must
FR-16	Export report	Approved case exports PDF summary, GeoJSON layers and candidate CSV with disclaimer and audit manifest.	Must
FR-17	Compare forcing/model runs	User can select two runs and inspect contour/score changes.	Should
FR-18	Record analyst decision	Candidate can be retained, downgraded or dismissed with signed note; model score is never silently overwritten.	Should

14 Non-Functional Requirements

ID	Quality attribute	Verifiable target for SIH demo
NFR-01	Reproducibility	Same inputs, seed, containers and model version reproduce equivalent artifacts; manifest contains hashes.
NFR-02	Performance	Stored demo case completes non-training processing in $\leq$10 min on the declared demo machine; UI interactions respond in $\leq$2 s for cached layers.
NFR-03	Reliability	Failed jobs are resumable by stage; no partially written artifact is marked complete.
NFR-04	Explainability	100% of ranked candidates expose all component scores, supporting spacetime point and AIS quality flags.
NFR-05	Auditability	Case creation, data changes, model runs, analyst reviews and exports are append-only events.
NFR-06	Security	Role-based access, encrypted transport, secrets outside source control, checksum validation and least-privilege services.
NFR-07	Accessibility	Keyboard operable controls, non-colour status labels, readable contrast and text alternative for map conclusions.
NFR-08	Portability	One documented container command starts API, worker, database and UI with the prepared demo assets.
NFR-09	Data integrity	All spatial objects carry CRS; all times are stored UTC; derived values retain provenance.
NFR-10	Honest uncertainty	No screen or export uses “culprit,” “guilty” or probability language without calibrated validation.

15 System Architecture

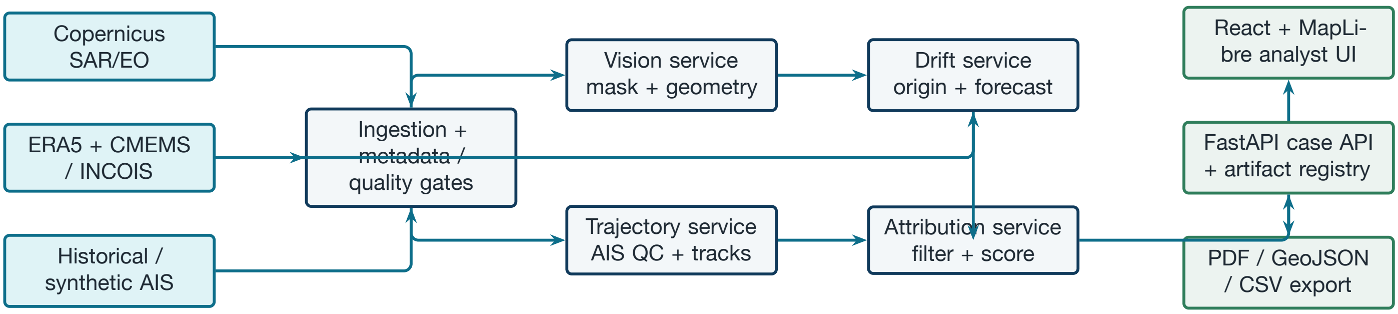

Figure 3: MVP logical architecture. Long-term platform services are not required to run the SIH demonstration.

15.1 Technology baseline

Layer	MVP choice	Reason / production path
Geospatial/ML	Python, PyTorch, Rasterio/GDAL, GeoPandas, xarray	Mature scientific stack; package versions pinned.
Trajectory model	OpenDrift/OpenOil	Open, scriptable, documented backward operation and oil module. Validate against PyGNOME case where feasible.
Data store	Parquet + DuckDB	Fast local analytics and simple packaging. Migrate spatial operational queries to PostGIS.
API/jobs	FastAPI + lightweight worker	Clear contracts; background stage execution; no Kubernetes dependency.
UI	React + MapLibre GL JS	Professional layered map, timeline and evidence panels. A Streamlit fallback may be retained only for internal debugging.
Packaging	Docker Compose	One-command reproducible demo; production may adopt managed orchestration later.

16 Processing Sequence and Module Contracts

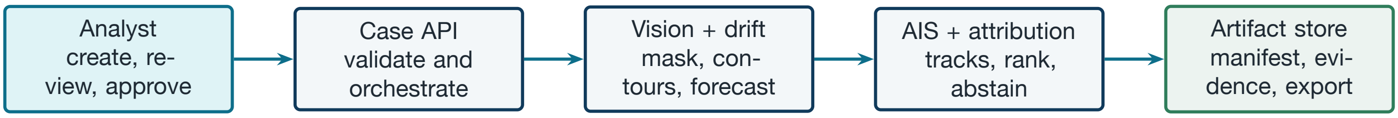

Figure 4: Case processing sequence. Each handoff writes a versioned artifact before the next stage begins.

16.1 Module outputs

Module	Required outputs	Failure output
Ingestion/QC	Canonical metadata, checksums, coverage report, immutable input refs.	Typed validation error and blocking quality flags.
Vision	Probability raster, mask GeoTIFF, slick GeoJSON, metrics JSON, preview tile.	No-detection or model-quality flag.
Drift	Time-indexed particle NetCDF/Parquet, density GeoTIFF, 50/90% contours, manifest.	Missing forcing, unstable run or invalid coastline response.
AIS	Clean observations, track segments, gap/quality flags, derived samples.	No coverage, invalid schema or no valid trajectories.
Attribution	Candidate table, component scores, evidence points, abstention code.	"Insufficient evidence" with machine-readable reasons.
Reporting	Human-readable report and geospatial/CSV bundle.	Export validation error; never a partial official bundle.

17 Data Model and API Contract

17.1 Core entities

Entity	Key fields	Notes
Case	case_id, AOI, start/end UTC, status, created_by	Parent object; state machine controls allowed actions.
SourceAsset	asset_id, URI, checksum, type, CRS, time, licence	Immutable after registration.
DetectionRun	run_id, model/version, config, quality, mask refs	Multiple runs allowed; one analyst-selected run.
Slick	slick_id, polygon, geometry metrics, review state	Geometry in WGS84 plus measurement CRS.
DriftRun	forcing IDs, ensemble parameters, seed, contours, horizon	Links to selected slick and source assets.
AISObservation	MMSI, UTC, lon/lat, SOG, COG, nav status, source	Original point is never overwritten.
TrackSegment	vessel key, interval, line geometry, gap/quality features	Derived points flagged and reproducible.
CandidateEvidence	vessel, score parts, closest event, explanation refs	Versioned by attribution-run ID.
AuditEvent	actor, action, UTC, before/after refs, reason	Append-only.

17.2 Representative endpoints

Endpoint	Method	Behaviour
`/cases`	POST	Create case with AOI and time window; return ID and validation status.
`/cases/{id}/assets`	POST	Register imagery, forcing or AIS asset; compute checksum and coverage.
`/cases/{id}/detect`	POST	Queue detection; return run ID.
`/detections/{run}/review`	POST	Accept/reject/uncertain with analyst reason.
`/cases/{id}/drift`	POST	Queue backward/forward ensemble with explicit config.
`/cases/{id}/candidates`	GET	Return ranked candidates, score parts and abstention state.
`/cases/{id}/export`	POST	Freeze signed report bundle after validation.
`/jobs/{id}`	GET	Return queued/running/succeeded/failed status and logs reference.

Listing 1: Minimal candidate response contract (illustrative JSON).


```json
{
  "case_id": "case_26143_demo_01",
  "status": "review_required",
  "score_type": "evidence_index",
  "candidates": [{
    "vessel_key": "mmsi:XXXXXXXXX",
    "score": 78.4,
    "components": {"space": 0.91, "time": 0.82,
                "forward_fit": 0.73, "behaviour": 0.31},
    "ais_quality": 0.94,
    "evidence_time_utc": "2026-08-20T04:15:00Z",
    "flags": ["short_gap_interpolated"]
  }],
  "abstention_reasons": []
}
```


18 Interface Specification

The main screen uses a 70/30 split: map on the left; case/evidence panel on the right. It must remain usable at 1366×768 without horizontal scrolling.

Region	Required content	Interaction
Top status bar	Case ID, acquisition time, state, selected run, data-quality badge.	Switch version; open manifest; never hide warnings.
Map layers	SAR, oil mask, origin contours, forward envelope, shoreline, AIS tracks.	Toggle, opacity, legend, click evidence; all layers time-labelled.
Timeline	Observation time, release window, vessel positions, forecast horizon.	Play/pause and scrub; UTC always visible.
Slick card	Geometry, detection confidence, wind/current coverage, review decision.	Accept/reject/uncertain with note.
Candidate list	Rank, score type, score parts, AIS quality, closest spacetime event.	Select track; compare candidates; dismiss with reason.
Evidence drawer	Charts/table for distance, origin density, speed/turn, gaps, forward-fit.	Link each claim to artifact and timestamp.
Export panel	Disclaimer, included artifacts, missing evidence and approver.	Export only after preflight passes.

19 Repository and Project Structure

Listing 2: Recommended mono-repository. Each service has a narrow responsibility.

```
oiltrace/
|-- apps/
| |-- web/ # React + MapLibre analyst interface
| `-- api/ # FastAPI routes and case state machine
|-- services/
| |-- ingestion/ # source adapters, checksums, metadata QC
| |-- vision/ # preprocessing, inference, vectorisation
| |-- drift/ # OpenDrift configs, ensembles, contours
| |-- ais/ # parsing, QC, segmentation, interpolation
| |-- attribution/ # gating, features, scoring, explanations
| `-- reporting/ # PDF/GeoJSON/CSV evidence bundle
|-- packages/
| |-- schemas/ # Pydantic/JSON schemas and shared types
| |-- geo/ # CRS, distances, raster/vector utilities
| `-- observability/ # structured logs, metrics, run manifests
|-- models/ # model cards and version manifests; no raw data
|-- configs/ # demo and environment configuration
```

```
|-- data/
| |-- demo/ # small redistribution-safe prepared case
| `-- manifests/ # source URLs, licences, checksums
|-- tests/
| |-- unit/ # equations, QC rules, schemas
| |-- integration/ # stage-to-stage contracts
| |-- scenarios/ # synthetic attribution benchmark
| `-- ui/ # critical browser journeys
|-- docs/ # architecture, methods, runbooks
|-- docker-compose.yml
|-- Makefile
`-- README.md
```

20 Security, Integrity and Responsible Use

- Roles: viewer, analyst, approver and administrator. Only approvers can freeze external reports.
- Secrets: tokens and credentials come from environment/secret store; never commit them or embed them in exported manifests.
- Integrity: checksum all input and derived artifacts; signed/frozen export manifest; append-only audit events.
- Privacy/security: restrict access to non-public AIS and vessel identity data; follow source terms; avoid public exposure of sensitive live tracks. IMO itself cautions that unrestricted AIS publication may affect ship and port security [12].
- Model abuse: do not label a vessel as guilty, trigger enforcement automatically, or conceal contradictory evidence.
- India context: Indian Coast Guard is the Central Coordinating Authority for oil-spill response under the national framework; the product is designed to support, not replace, authorised agencies and established response processes [21]. MARPOL Annex I remains the international prevention framework for oil pollution from ships [22].

21 Test Strategy and Acceptance Gate

Test level	Coverage	Release gate
Unit	CRS/time conversion, geometry, trajectory feasibility, score bounds, contour generation, state transitions.	All deterministic tests pass; seeded stochastic tests meet tolerances.
Contract	Every service request/response and artifact schema.	No undocumented breaking field; invalid inputs fail clearly.
Integration	Prepared imagery → mask → drift → AIS → ranking → export.	Clean run from empty local environment; artifact manifest complete.
Scientific regression	Fixed held-out scenes and synthetic cases.	Metrics do not regress beyond predeclared tolerance.
UI/browser	Create case, review mask, inspect candidate, abstain/approve, export.	No overlap/clipping at target viewport; keyboard path works.
Adversarial	Look-alike, wrong forcing, no AIS, gaps, dense traffic, two modes.	Correct warnings/abstention; no unsupported accusation.

22 SIH Demonstration Scenario

Prepared case: a georeferenced Sentinel-1 slick scene or a redistribution-safe derived tile; matched environmental history; real sample AIS where licensing and alignment permit, otherwise a clearly labelled synthetic traffic field containing normal vessels, a plausible source vessel, a coverage gap, and a distractor passing near the observed slick but inconsistent with the inferred release time.

Five-minute narrative

1. Open case and show source provenance plus current data-quality status.
2. Run or load U-Net inference; compare raw SAR, probability and mask; accept the slick.
3. Show measured geometry and explain why thickness is not claimed.
4. Animate backward ensemble to 50/90% origin contours; switch to the forward 24-hour envelope.
5. Overlay AIS history; compare the proximity-only distractor with the drift-consistent candidate.
6. Open the score decomposition and AIS gap warning; deliberately show the score as an index.
7. Export the lead report, then open a no-AIS case to demonstrate honest abstention.

Evidence to show judges: held-out segmentation metrics; source-contour coverage on synthetic releases; Top-1/Top-3 comparison versus proximity-only; one ablation; a complete manifest; and one failure case. Avoid live network dependencies during the primary demo.

23 Implementation Roadmap

Table 5: Relative plan; dates should be aligned to the official SIH 2026 calendar.

Sprint	Objective	Deliverables	Exit evidence
S0: Freeze	Reproducible data contract and demo case	Repository, manifests, environment, synthetic generator, baseline metrics	Clean setup and immutable input hashes
S1: Detect	Reliable candidate mask and geometry	U-Net baseline, look-alike class, review UI, vector outputs	Held-out metrics and visual QA
S2: Transport	Probabilistic hindcast/forecast	Forcing adapter, ensemble runner, contours and sensitivity	Source coverage and displacement metrics
S3: Correlate	AIS processing and transparent ranking	QC, segments, candidates, score breakdown, abstention	Synthetic Top-k and baseline ablation
S4: Product	Integrated analyst workflow	Map/timeline, audit log, export bundle, accessibility pass	Browser journey and complete manifest
S5: Harden	Judge-ready reliability	Cached demo, failure cases, performance pass, video backup	Two successful cold runs on demo machine
Post-SIH	Validate and scale	Real case partnerships, calibration set, PostGIS, regional forcing, operational security review	Domain-owner acceptance and monitored pilot

24 Proposed Team Workstream Allocation

The roles below are a delivery proposal, not a claim about each member's prior expertise. Team lead retains integration and scope authority.

Member	Primary workstream	Concrete ownership
Sourav Kumar (Lead)	Architecture, integration and demo	Scope, interfaces, release gate, final narrative and contingency plan.
Aashi Soni	Research, data governance and evaluation	Literature ledger, dataset cards, licensing, metric protocol and report evidence.
Anish Mall	SAR/EO machine learning	Preprocessing, U-Net training/inference, uncertainty and geometry extraction.
Ayush Kumar Sharma	Drift and environmental data	ERA5, CMEMS and INCOIS adapters; OpenDrift ensembles, contours and sensitivity runs.
Ayush Kesarwani	AIS and attribution	AIS QC/reconstruction, candidate generation, scoring, synthetic scenarios and ablations.
Devansh Goenka	Product, UI and deployment	FastAPI integration; React and MapLibre workflow; export, Docker and browser QA.

24.1 RACI for critical outputs

Deliverable	Sourav	Aashi	Anish	A. Sharma	A. Kesarwani	Devansh
Research/evidence ledger	A	R	C	C	C	I
Detection model	A	C	R	I	I	C
Drift engine	A	C	C	R	C	I
AIS attribution	A	C	I	C	R	C
UI and deployment	A	C	C	C	C	R
Final demo/export	A/R	C	C	C	C	R

R = Responsible, A = Accountable, C = Consulted, I = Informed.

25 Risks, Dependencies and Mitigations

Table 6: Top project risks.

Risk	Likelihood	Impact	Mitigation / trigger
No matched real AIS for spill region/time	High	High	Use licensed US sample or synthetic aligned tracks; label evidence type prominently; never imply India-ready data access.
Detection domain shift	High	High	Scene-level holdout, look-alike negatives, quality gate and analyst review; show failure case.
Environmental field too coarse	Medium	High	Compare products/perturbations, widen contours, restrict horizon and abstain near unresolved coastlines.
Backward inference overconfidence	Medium	High	Ensembles, multimodal contours, release-window sweep and forward consistency check.
AIS gaps misinterpreted	High	High	Separate reception coverage from onboard behaviour; reliability multiplier and neutral language.
Compute/network failure during judging	Medium	High	Cache approved demo artifacts; support deterministic replay; keep a recorded backup.
LaTeX/report or UI overflow	Low	Medium	Automated build plus rendered visual review at target viewport and all PDF pages.
Scope inflation into SAM/GNN/Kubernetes	Medium	Medium	Keep research backlog separated; enforce Must-only release gate.
Unsupported probability claim	Medium	High	Display evidence index until Brier/ECE calibration gate passes.
Sensitive data exposure	Low/Medium	High	Least privilege, access controls, redacted demo identifiers and export preflight.

26 Future Research Roadmap

Only after the MVP passes its component and end-to-end gates:

1. Multimodal SAR/optical late fusion where acquisitions are sufficiently close in time.
2. Regional high-resolution currents, tides and shoreline interaction for ports and nearshore cases.
3. Empirical calibration using independently verified vessel–spill cases.
4. Learned trajectory anomaly models only after a representative “normal behaviour” corpus exists.
5. FTLE/coherent-structure analysis as a transport diagnostic, not as an automatic confidence generator.
6. Radar/RF/optical ship detections to cover non-reporting vessels, subject to authorised data access.
7. PostGIS/object-store migration, scheduled ingestion, model monitoring and operational security assessment.

27 Final Acceptance Checklist

	Gate	Evidence
☐	Problem ID 26143 and NTRO scope are correctly represented.	Cover and scope review
☐	Every dataset has source, licence/access note, geography and limitation.	Dataset cards/manifests
☐	Train/validation/test separation is by scene/event, not random overlapping tiles.	Split manifest
☐	Detection, drift and attribution metrics are reported separately.	Evaluation report
☐	One proximity-only baseline and one ablation are shown.	Result table/plot
☐	Candidate screen shows score type, components and AIS quality.	Browser test
☐	“No candidate” and “insufficient evidence” cases pass.	Scenario tests
☐	Report uses lead-generation disclaimer and contains complete provenance.	Export preflight
☐	Demo runs without live external services and has a recorded fallback.	Cold-run checklist
☐	PDF and UI have no clipped, overlapping or overflowing content.	Rendered visual QA

28 Conclusion

OilTrace is technically credible when presented as an uncertainty-aware investigation assistant, not an autonomous accusation engine. The winning SIH narrative is an end-to-end chain with measurable behaviour: robust candidate detection, probabilistic origin inference, physically consistent AIS correlation, clear abstention, and an evidence interface that lets an authorised analyst challenge every step. This is both more defensible and more operationally useful than adding unvalidated advanced models to an incomplete pipeline.

Verified References

[1] European Maritime Safety Agency. "CleanSeaNet: Satellite-based services for oil spill and vessel detection." (2026), [Online]. Available: https://emsa.europa.eu/csn-menu.html (visited on 08/27/2026).

[2] NOAA Office of Response and Restoration. "WebGNOME introduction." (2026), [Online]. Available: https://gnome.orr.noaa.gov/doc/introduction.html (visited on 08/27/2026).

[3] K.-F. Dagestad, J. Röhrs, O. Breivik, and B. Ådlandsvik, "OpenDrift v1.0: A generic framework for trajectory modelling," *Geoscientific Model Development*, vol. 11, pp. 1405–1420, 2018. DOI: 10.5194/gmd-11-1405-2018. [Online]. Available: https://doi.org/10.5194/gmd-11-1405-2018.

[4] OpenDrift Project. "Introduction to OpenDrift." (2026), [Online]. Available: https://opendrift.github.io/ (visited on 08/27/2026).

[5] D. Luo, P. Chen, J. Yang, X. Chen, X. Li, G. Zheng, L. Ren, and Y. Zhao, "A new ship tracing technology from oil spills based on multi-source data," *Marine Pollution Bulletin*, vol. 205, p. 116 808, 2024. DOI: 10.1016/j.marpolbul.2024.116808. [Online]. Available: https://doi.org/10.1016/j.marpolbul.2024.116808.

[6] Copernicus Data Space Ecosystem. "Sentinel-1 mission documentation." (2026), [Online]. Available: https://documentation.dataspace.copernicus.eu/Data/SentinelMissions/Sentinel1.html (visited on 08/27/2026).

[7] Copernicus Data Space Ecosystem. "Sentinel-1 GRD documentation and processing chain." (2026), [Online]. Available: https://documentation.dataspace.copernicus.eu/APIs/SentinelHub/Data/S1GRD.html (visited on 08/27/2026).

[8] C. Brekke and A. H. S. Solberg, "Oil spill detection by satellite remote sensing," *Remote Sensing of Environment*, vol. 95, no. 1, pp. 1–13, 2005. DOI: 10.1016/j.rse.2004.11.015. [Online]. Available: https://doi.org/10.1016/j.rse.2004.11.015.

[9] "Oil spill detection from synthetic aperture radar earth observations: A meta-analysis and comprehensive review," *GIScience & Remote Sensing*, 2021. DOI: 10.1080/15481603.2021.1952542. [Online]. Available: https://doi.org/10.1080/15481603.2021.1952542.

[10] J. Röhrs, K.-F. Dagestad, H. Asbjørnsen, T. Nordam, J. Skancke, C. E. Jones, and C. Brekke, "The effect of vertical mixing on the horizontal drift of oil spills," *Ocean Science*, vol. 14, pp. 1581–1601, 2018. DOI: 10.5194/os-14-1581-2018. [Online]. Available: https://doi.org/10.5194/os-14-1581-2018.

[11] NOAA Office of Response and Restoration. "PyGNOME introduction and documentation." (2026), [Online]. Available: https://gnome.orr.noaa.gov/doc/pygnome/introduction.html (visited on 08/27/2026).

[12] International Maritime Organization. "AIS transponders and carriage requirements." (2026), [Online]. Available: https://www.imo.org/en/ourwork/safety/pages/ais.aspx (visited on 08/27/2026).

[13] BOEM, NOAA and U.S. Coast Guard. "AccessAIS – marinecadastre.gov." (2026), [Online]. Available: https://marinecadastre.gov/accessais/ (visited on 08/27/2026).

[14] R. Trujillo-Acatitla, J. Tuxpan-Vargas, C. Ovando-Vázquez, and E. Monterrubio-Martínez, *Sentinel-1 sar oil spill image dataset for train, validate, and test deep learning models. part i*, 2024. DOI: 10.5281/zenodo.8346860. [Online]. Available: https://doi.org/10.5281/zenodo.8346860.

[15] R. Trujillo-Acatitla, J. Tuxpan-Vargas, C. Ovando-Vázquez, and E. Monterrubio-Martínez, *Sentinel-1 sar oil spill image dataset for train, validate, and test deep learning models. part ii*, 2024. DOI: 10.5281/zenodo.8253899. [Online]. Available: https://doi.org/10.5281/zenodo.8253899.

[16] R. Trujillo-Acatitla, J. Tuxpan-Vargas, C. Ovando-Vázquez, and E. Monterrubio-Martínez, *Sentinel-1 sar oil spill image dataset for train, validate, and test deep learning models. part iii*, 2024. DOI: 10.5281/zenodo.13761290. [Online]. Available: https://doi.org/10.5281/zenodo.13761290.

[17] M. Krestenitis, G. Orfanidis, K. Ioannidis, K. Avgerinakis, S. Vrochidis, and I. Kompatsiaris, "Oil spill identification from satellite images using deep neural networks," *Remote Sensing*, vol. 11, no. 15, p. 1762, 2019. DOI: 10.3390/rs11151762. [Online]. Available: https://doi.org/10.3390/rs11151762.

[18] Copernicus Marine Service. "Global ocean physics analysis and forecast: GLOBAL_ANALYSISFORECAST_PHY_001_024." (2026), [Online]. Available: https://data.marine.copernicus.eu/product/GLOBAL_ANALYSISFORECAST_PHY_001_024/description (visited on 08/27/2026).

[19] Indian National Centre for Ocean Information Services. "Online oil spill advisory system (OOSA)." (2026), [Online]. Available: https://oosa.incois.gov.in/OOSA/ (visited on 08/27/2026).

[20] F. M. Bianchi, M. M. Espeseth, and N. Borch, "Large-scale detection and categorization of oil spills from SAR images with deep learning," *Remote Sensing*, vol. 12, no. 14, p. 2260, 2020. DOI: 10.3390/rs12142260. [Online]. Available: https://doi.org/10.3390/rs12142260.

[21] Indian Coast Guard. "Marine environment and national oil spill disaster contingency plan." (2024), [Online]. Available: https://www.indiancoastguard.gov.in/marine-environment (visited on 08/27/2026).

[22] International Maritime Organization. "International convention for the prevention of pollution from ships (MARPOL)." (2026), [Online]. Available: https://www.imo.org/en/about/conventions/p.s/international-convention-for-the-prevention-of-pollution-from-ships-%28marpol%29.aspx (visited on 08/27/2026).